

Emily Zhou

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EDUCATION

University of Virginia

May 2026

Bachelor of Arts in Computer Science, Minor in Data Science, and Minor in French

GPA: 3.89

Relevant Coursework: Data Structures and Algorithms, Machine Learning, Software Engineering, Discrete Math and Theory, Cloud Computing, Computer Systems and Organization, Cybersecurity, Database Systems, Algorithmic Economics

TECHNICAL SKILLS

Languages: Java, Python, HTML/CSS, Javascript, Go, C

Tools & IDEs: YOLOv8, Jenkins, Jira, Bootstrap, Django, JavaFX, Figma, Pandas, Tensorflow, SciKitLearn, Angular

Database & Cloud: PostgreSQL, MySQL, AWS EC2, AWS RDS, AWS S3, Heroku, Docker, Kubernetes, DBeaver, MongoDB

Certificates: AWS Certified Cloud Practitioner

WORK EXPERIENCE

Capital One TDP | *Software Engineer*

Richmond, VA (Aug 2026)

Business and AI Institute, Rewriting the Code | *Project Lead*

Charlottesville, VA (Jan 2025 - May 2026)

- Built a Slackbot using LangGraph + Together AI (Mistral) to auto-answer FAQs for 40,000+ members.
- Stored context in PostgreSQL (EC2) + ChromaDB (Docker) for vector search and real-time retrieval.
- Developed React web admin tool for nontechnical RTC staff to manage message data and escalate unresolved questions.
- Integrated Slack reaction feedback loop; pilot testing showed ~85% positive response satisfaction.

Tools: Python, LangGraph, LLM, Mistral, TogetherAI, PostgreSQL, AWS EC2, Docker, ChromaDB, Slack API, Ngrok

CACI, Customs and Border Protection (DHS) | *Software Developer Intern*

Ashburn, VA (May 2025 - Aug 2025)

- Revamped legacy Angular forms by implementing 40+ dynamic fields and reusable components, reducing frontend tech debt.
- Built and tested RESTful APIs using Spring Boot, Postman, and Swagger; authored 100+ JUnit tests to validate API functionality and improve backend reliability.

Tools: PostgreSQL, Java, Spring Boot, Maven, Angular (JavaScript), JUnit, Swagger, Docker, Kubernetes, GitLab, Jira

Centrifuge LLC, Customs and Border Protection (DHS) | *AI/CV Intern*

Reston, VA (Jun 2024 - Aug 2024)

- Developed real-time object detection using YOLOv8 on AWS EC2, processing thousands of live IP camera frames daily.
- Implemented multithreaded processing for 5× faster throughput, cutting analysis delays for 100k+ tracked events.
- Stored detections in PostgreSQL and AWS S3, managing, with fully containerized deployment via Docker and Kafka.
- Streamlined visual data pipelines, improving incident response efficiency through real-time automated monitoring.

Tools: Python, YOLOv8, AWS EC2, AWS S3, PostgreSQL, Docker, GitHub, Bash, DBeaver, Kafka, Colab

PROJECTS

ML4VA NOVA Car Crash Hotspot Severity Prediction

Fall 2024

- Co-developed a machine learning pipeline to identify high-risk roadways in Northern Virginia using traffic crash data from VDOT, targeting safety and infrastructure planning using random forest and neural networks.
- Trained and optimized Random Forest models to predict crash severity (RMSE: 0.94) and classify road defects (F1-score: 0.96), significantly outperforming the neural network baseline.

Tools: Python, Pandas, scikit-learn, Tensorflow, Random Forest Classifier/Regressor, Matplotlib, Seaborn

Hoos Reporting Honor | *Product Manager*

Spring 2024

- Led 5-person Agile team to build a Django-based honor reporting application for UVA, hosted on Heroku.
- Designed PostgreSQL data models, Bootstrap front-end, and integrated AWS S3 for scalable file storage.
- Automated CI/CD pipelines on Heroku with seamless GitHub integration, enhancing deployment efficiency.

Tools: Python, Django, Heroku, AWS S3, PostgreSQL, Bootstrap, GitHub, Figma, HTML, CSS

LEADERSHIP & VOLUNTEER EXPERIENCE

CS 4750 Database Systems | *Teaching Assistant*

Jan 2026 - Present

- Supported 200+ students in learning core database systems concepts through office hours, grading, and technical guidance.

CS 3240 Software Engineering | *Teaching Assistant*

Aug 2024 - Present

- Mentored 4+ student developer teams through Agile sprints on full-stack web dev and software engineering principles.

Costuming, Cosplay, and Crafts, UVA CIO | *Social Media Manager*

Jan 2024 - May 2025

- Designed Instagram marketing for 200+ members and acquired \$10k+ of UVA Library, UVA Arts, and UVA Parents funding.